

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000785	Version:	4.0, CURRENT
	Status:	Effective	Effective Date:	23-May-2011
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DG-57 / DG-517 Service Manual

USER MAINTENANCE

Procedure description
This procedure describes how to perform user maintenance on the System. 1. How to refill the Wash Solutions Bottle and empty the Waste Solution Bottle. Daily, before starting to work, a revision of bottles' level (wash solutions and waste) must be done. During working, if a bottle refill/empty is required the system warns to user. 2. How to empty the waste tray. Daily, before starting to work, waste tray must be done. During working, if emptying the waste tray is required the system warns to user. 3. How to do a Final Washing. At the end of daily work, a final washing is required in order to keep the fluidic system in clean conditions. 4. How to do a general cleaning of the interior and exterior of equipment. Daily, a check of interior and exterior of equipment must be done and cleaning if required. 5. How to do a Rinse. Monthly or whenever the equipment is moved from its working area.

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SETUP

1. PC turned on.
2. Equipment turned off.

	DANGER! ELECTRICAL DISCHARGE Although this instrument is completely isolated and connected to earth, it is important that all users realize the danger of using liquids near an electric mains connection. In the case of a liquid spill, the instrument should be immediately disconnected from the electric mains supply and must not be reconnected until it has been inspected by the Technical Service.
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	DANGER! ENVIRONMENTAL CONTAMINATION If there has been a spill of any substance that is a danger to the environment, the instrument should be immediately disconnected from the electric mains supply and isolated from the work area. Contact the authorized Technical Service and explain the dangers associated with the spilt dangerous substances.
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	DANGER! DANGEROUS WASTE SUBSTANCES The contents of the waste bottle should be emptied into special containers provided for this type of waste that must be eliminated in a selective manner.
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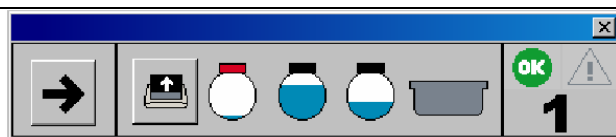
EQUIPMENT

1. Concentrate of Wash Solutions A & B.
2. Distiller water.
3. Ethylic alcohol 70%.

PROCEDURE 1

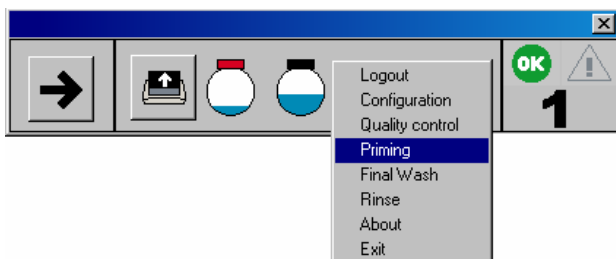
Process	Expected Results
How to refill the Wash Solutions Bottles 1. Make sure the equipment is turned off and the door is opened. 2. Unplug racors of Wash Solution Bottle A. 3. Open the cap and remove it. 4. Refill the Bottle A as the Instruction For Use of Concentrate Wash Solution A indicates. 5. Plug the racors of Wash Solution Bottle A, close the cap and place the bottle properly to assure the correct position. 6. Unplug racors of Wash Solution Bottle B. 7. Open the cap and remove it. 8. Refill the Bottle B as the Instruction For Use of Concentrate Wash Solution B indicates. 9. Plug the racors of Wash Solution Bottle B, close the cap and place the bottle properly to assure the correct position. 10. Close the door and turn the equipment on. 11. Run User Software System.	--- --- --- --- --- --- --- --- --- --- New window appears: (Picture 1)

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Picture 1

12. On main window, click on secondary mouse button and select *Priming* option.



Picture 2

The system starts Priming.

How to empty the Waste Solution Bottle

13. Unplug racors of Waste Solution Bottle.
14. Open the cap and remove it.
15. Empty the bottle content in a proper container, according to the current legislation of your country.
16. Plug the racors of Waste Solution Bottle, close the cap and place the bottle properly to assure the correct position.
17. Close the door and turn the equipment on.

PROCEDURE 2

Process	Expected Results
How to empty the waste tray	
1. Open instrument doors.	---
2. Remove the waste tray.	---
3. Empty the waste tray content in a proper container, according to the current legislation of your country.	---
4. If required, clean the waste tray with a damp cloth and soap.	---
5. Additionally, alcohol may be used to clean and disinfect.	---

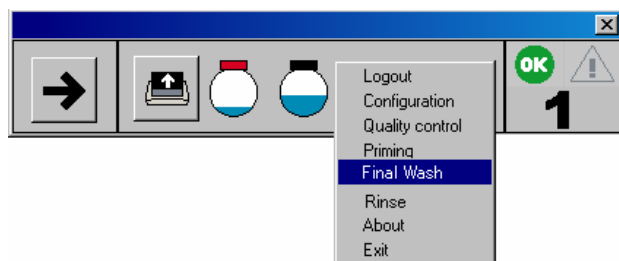
PROCEDURE 3

Process	Expected Results
How to do a Final Washing	
1. Make sure the equipment is turned on and the door is closed.	---
2. Run the User Software System.	
Picture 3	
3. On main window, click on secondary mouse button and select <i>Final wash</i>	The system starts

New window appears:
(Picture 5)

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option.



Picture 4

Priming.

- When turning off the instrument and software at the end of daily work, the software will prompt the user to perform a Final Wash. By clicking Yes the Final Wash will be performed automatically.

PROCEDURE 4

Process	Expected Results
How to do a general cleaning of the interior and exterior of equipment	
1. Make sure the equipment is turned off and disconnected from the electric mains supply.	---
2. Check the interior and exterior surfaces of equipment.	---
3. If required, clean using a damp cloth with distiller water.	---
4. Check the bottles.	---
5. If required, the bottles should be removed from equipment and washed externally with soap and water.	---
6. Check the samples and reagents carousel.	---
7. If required, the carousel may be removed in order to clean it. It should be cleaned with a damp cloth and soap.	---
8. Additionally to distilled water and soap, alcohol may be used to clean and disinfect the area that has been in contact with blood or other biological contaminant.	---
9. Check the front glass.	---
10. If required, clean the front glass always with mild and not abrasive products in order to avoid any scratch to the glass during the cleaning process.	---

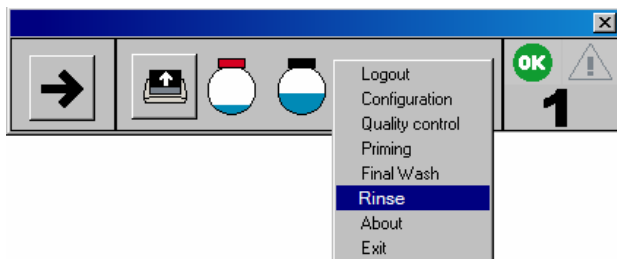
PROCEDURE 5

Process	Expected Results
How to do a Rinse	
1. Make sure the equipment is turned on and the door is closed.	---
2. Run the User Software System.	---
 A screenshot of a software interface. On the left, there are five icons: a right-pointing arrow, a document with an upward arrow, a red-topped bottle, a blue-topped bottle, and a grey tray. To the right of these icons is a menu with the following options: Logout, Configuration, Quality control, Priming, Final Wash, Rinse (highlighted in blue), About, and Exit. Further right, there is a green 'OK' button and a warning icon (triangle with an exclamation mark) next to a large number '1'.	---

Picture 5

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3. On main window, click on secondary mouse button and select *Rinse* option.



Picture 6

4. The software shall display the following question: "This procedure can't detect the proper racors connection. Have you checked it?" "Yes/No".
Check the racor connections and click in "Yes".
5. When the program finishes the first part of the Rinse procedure, display the following warning/message: "Place containers/bottles filled with Sol.A and Sol.B".
6. Click on "Yes"
7. The program finishes the Rinse procedure.

DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
2.0	2005/05/09	NA	Document REGD-0000785_v2 replaces document 57012V0M000
3.0	2010/09/10	- Updated Pictures 1 to 6. - Added the procedure 5: How to do a Rinse. New functionality.	Document update due to the Rinse procedure implemented.
4.0	See document header	Updated document change history: -Date format changed to YYYY/MM/DD. -Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
4.0	See document header	Document REGD-0000785_v4 replaces document REGD-0000785_v3.	Document version changed.
4.0	See document header	Document title.	Commercial reference substituted by project reference.
4.0	See document header	Document header.	Increase information to easy document search.